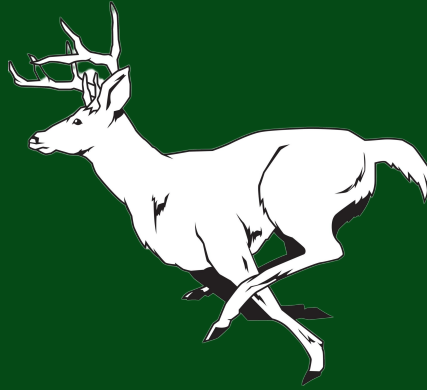




Preliminary Design Review

Deer Dash – Wolf Park Carcass Feeding



Project Manager,
Stakeholder Liaison



Design Lead



Analysis Lead



Manufacturing Lead,
Purchasing/Econ Lead



Testing and
Validation Lead



Documentation and
Communication Lead

Problem Statement



Wolf Park needs to safely deliver animal carcasses into wolf enclosures to allow the wolves to feed with more natural behaviors like leaping and taking down their food.



Wolves' leaping behavior can be hard to mimic in captivity due to their large natural prey



Sparrow from Wolf Park



Impact

- Improve enrichment for wolves
- Improve stimulation of wolves during feeding
- Increase safety for employees
- Improve entertainment for guests



Timber



Sparrow



Aspen



Máni

Customer Requirements

Facility

- Tall enough to go over enclosure fence
- Extend into middle of enclosure (~20 ft)
- Outside 24/7
- Away from safety fence

Replicable

- Simple to construct
- Economical
- Wholly mechanical

Enrichment

- Wolves can safely bite the carcass
- Carcasses within jumping height of wolves



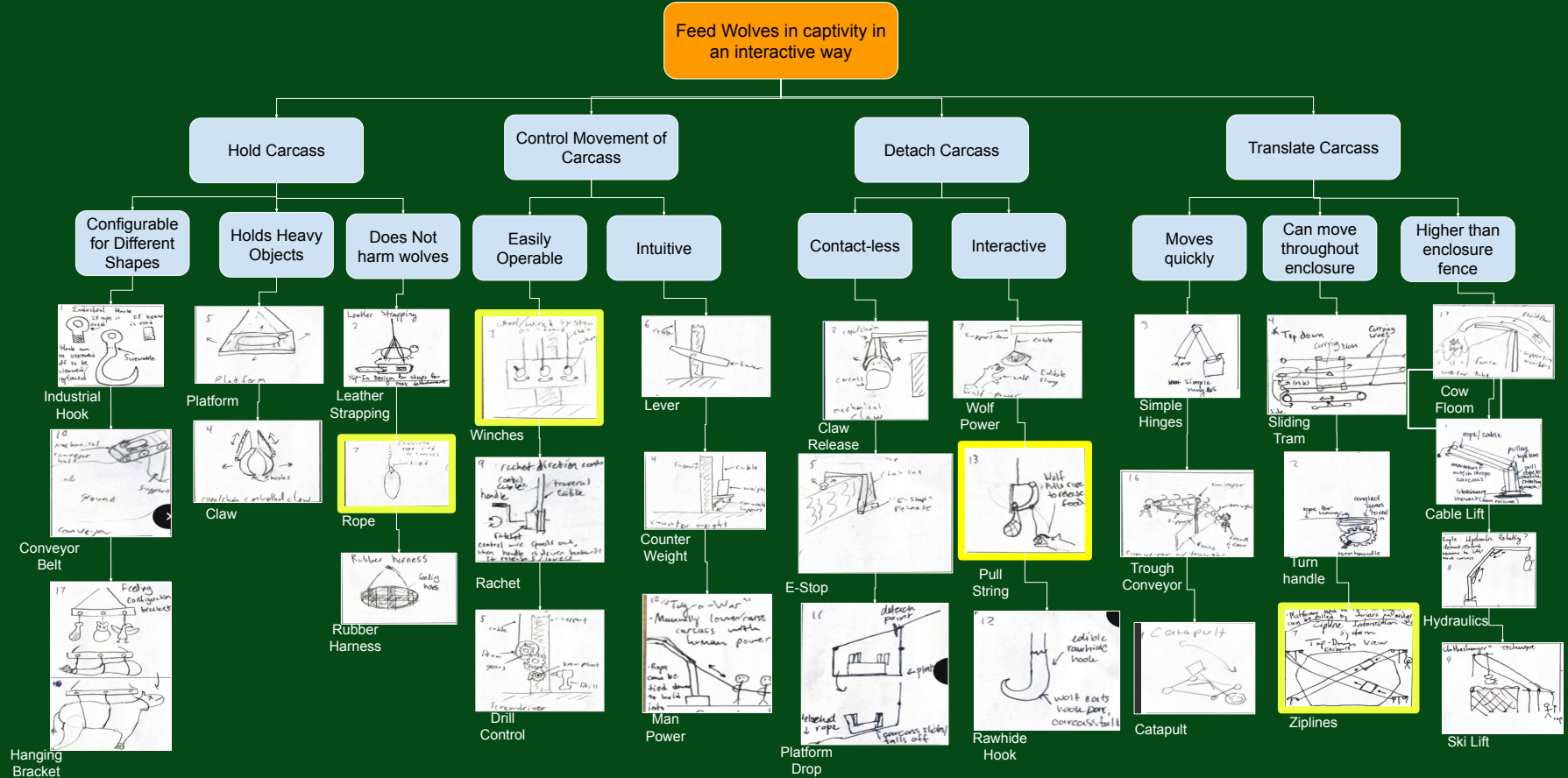
Engineering Requirements

	Engineering Requirement	Customer Need it Addresses	Metric / Target	Pass / Fail Criteria	Verification Method
	Load Capacity	Wolves pull downward on deer and calf carcasses	Carries ≥ 200 lbs and can handle loads ≥ 500 lbs	F.O.S of 2	FEA
	Material Choice	Outside 24/7	$>90\%$ of materials used do not rust or decay in weather	Fails if materials damaged by weather and rust	Material Properties
	Simple Design	Easily reproducible	<4 moving parts, manufactured with $>80\%$ OEM parts	Fails if >4 moving systems	CAD and BOM
	Height	Goes over enclosure fencing	>17 ft	Within dimension requirements	CAD verification

Engineering Requirements

	Engineering Requirement	Customer Need it Addresses	Metric / Target	Pass / Fail Criteria	Verification Method
	Span of device into enclosure	Reaches far into enclosure	≥ 20 ft	Passes if within dimension requirements	CAD verification
	Height of hook mechanism	Wolves able to jump at carcass	Hangs 6-8 ft from ground	Fails if below 6 ft or stuck above 8 ft	CAD verification
	Hardness of wolf-device	Wolves can safely bite carcass	Softer than enamel ($\leq$ Shore Hardness 70)	Fails if ≥ 80	Shore Hardness Test
	Distance from fence	Away from safety fence	≥ 3 ft from fence	Fails structure is within 3 ft of fence	CAD verification

Functional Decomposition



Sample Calculations - Tension Forces

Assumptions for minimum specification:

- Post Height (H) is 17ft
- Height of carcass from ground (h) is 8ft
- Carcass is 3.5ft long deer leg
- Distance between support beams is 20ft
- Total force applied is 500 lbf in the center of the wire
- Point of carcass attachment is on the tension line
- Carcass is at static equilibrium

Vertical displacement of load attachment point:

$$X = 17\text{ft} - 8\text{ft} - 3.5\text{ft} = 5.5\text{ft}$$

Therefore:

$$\theta = \tan^{-1}(5.5/10) = 28.8^\circ$$

Tension force calculation:

$$\sum F_y = 0 = T_1 \sin(\theta) + T_2 \sin(\theta) - F_{\text{load}}$$

$$2 * T \sin(28.8^\circ) = 500\text{lb}$$

$$T = T_1 = T_2 = 519\text{ lbf}$$

$$\text{Minimum Line Weight} = 1040\text{ lbf with factor of safety } 2$$

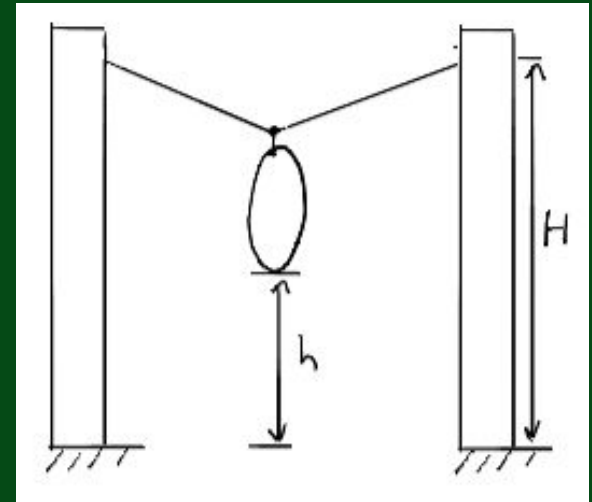
Beam Bending Moment:

$$M = T \cos(\theta) * H$$

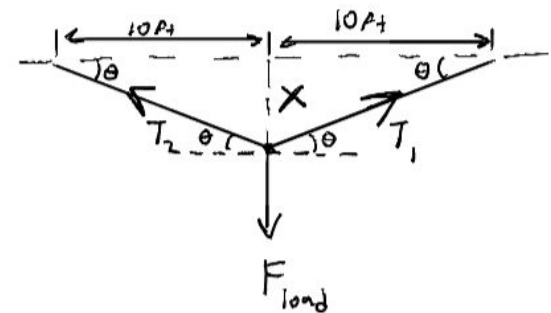
$$= (519\text{ lbf})$$

$$* \cos(28.8^\circ) * (17\text{ft})$$

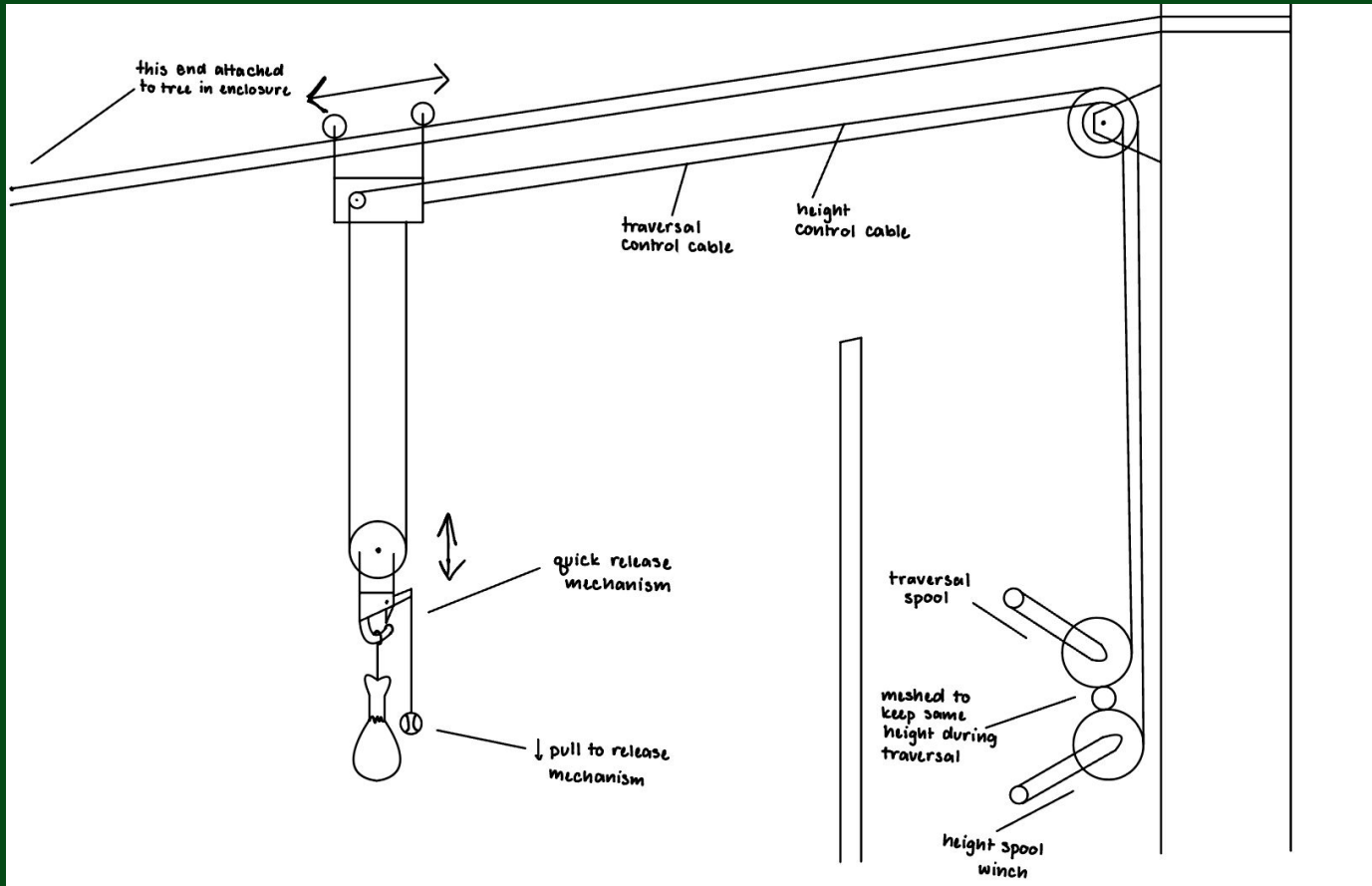
$$M_{\text{min}} = 7732\text{ lbf-ft}$$



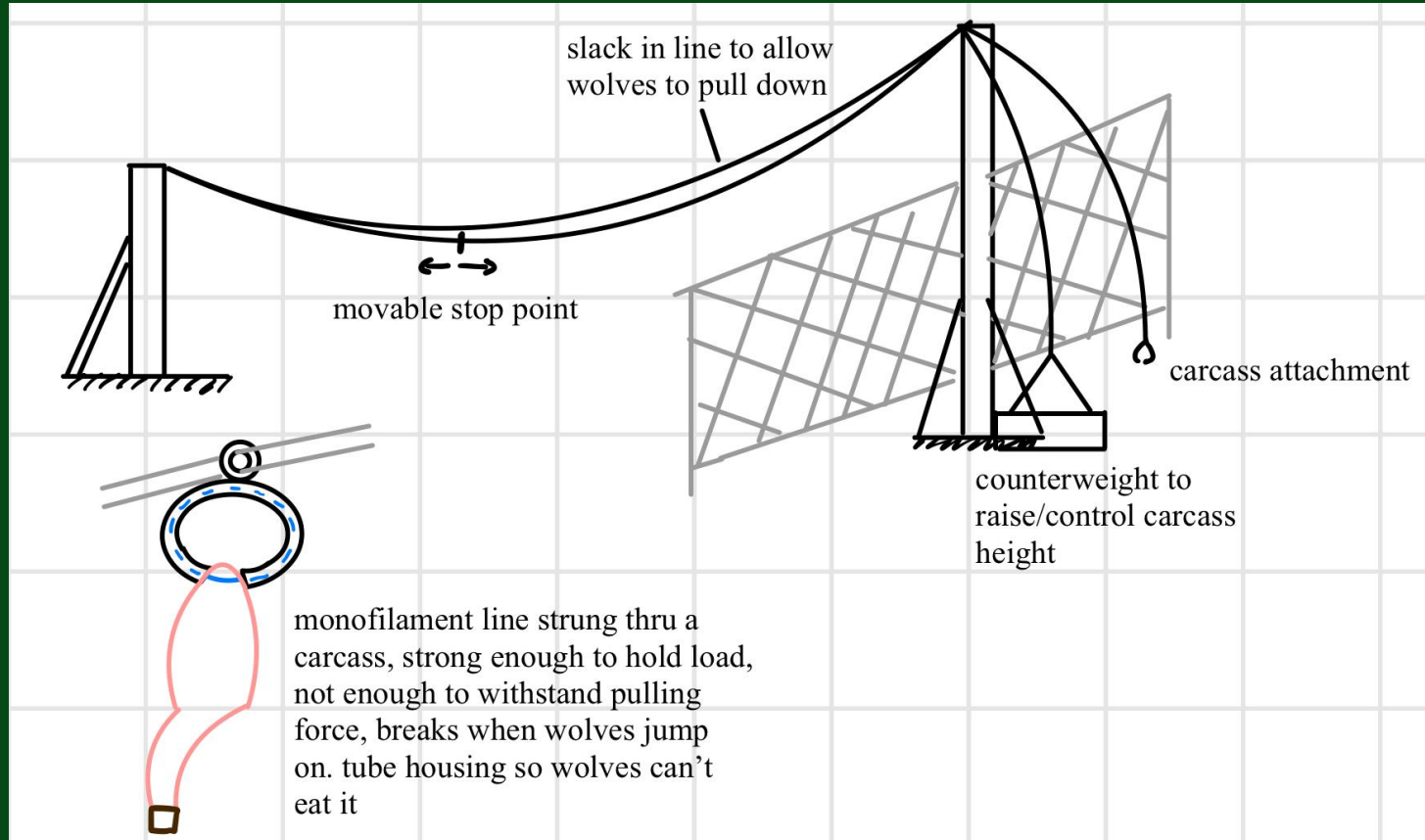
FBD



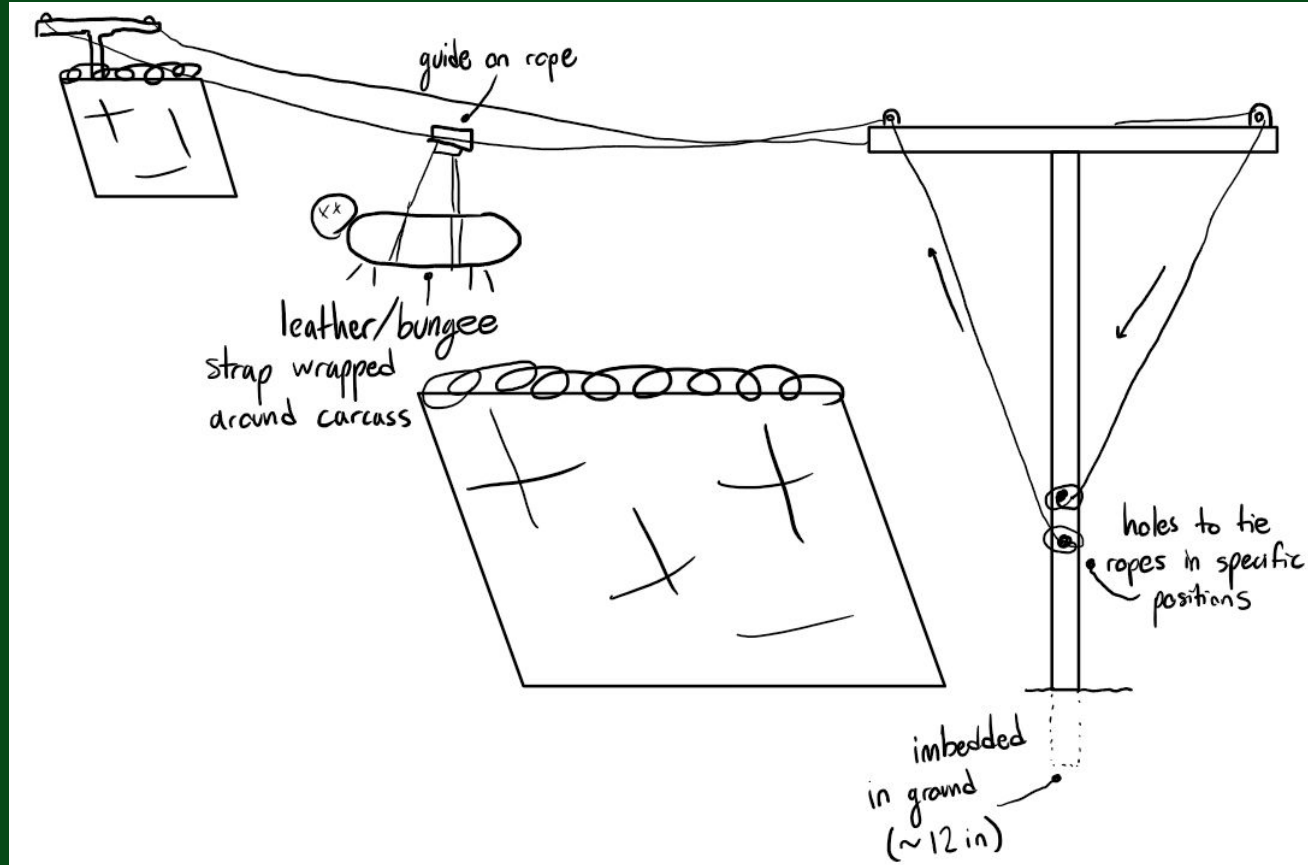
Concept 1 – Meshed Gears with Cables



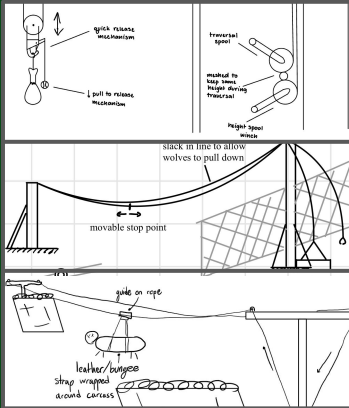
Concept 2 – Counterweight



Concept 3 – Manual Ski Lift Cable



Concept Selection



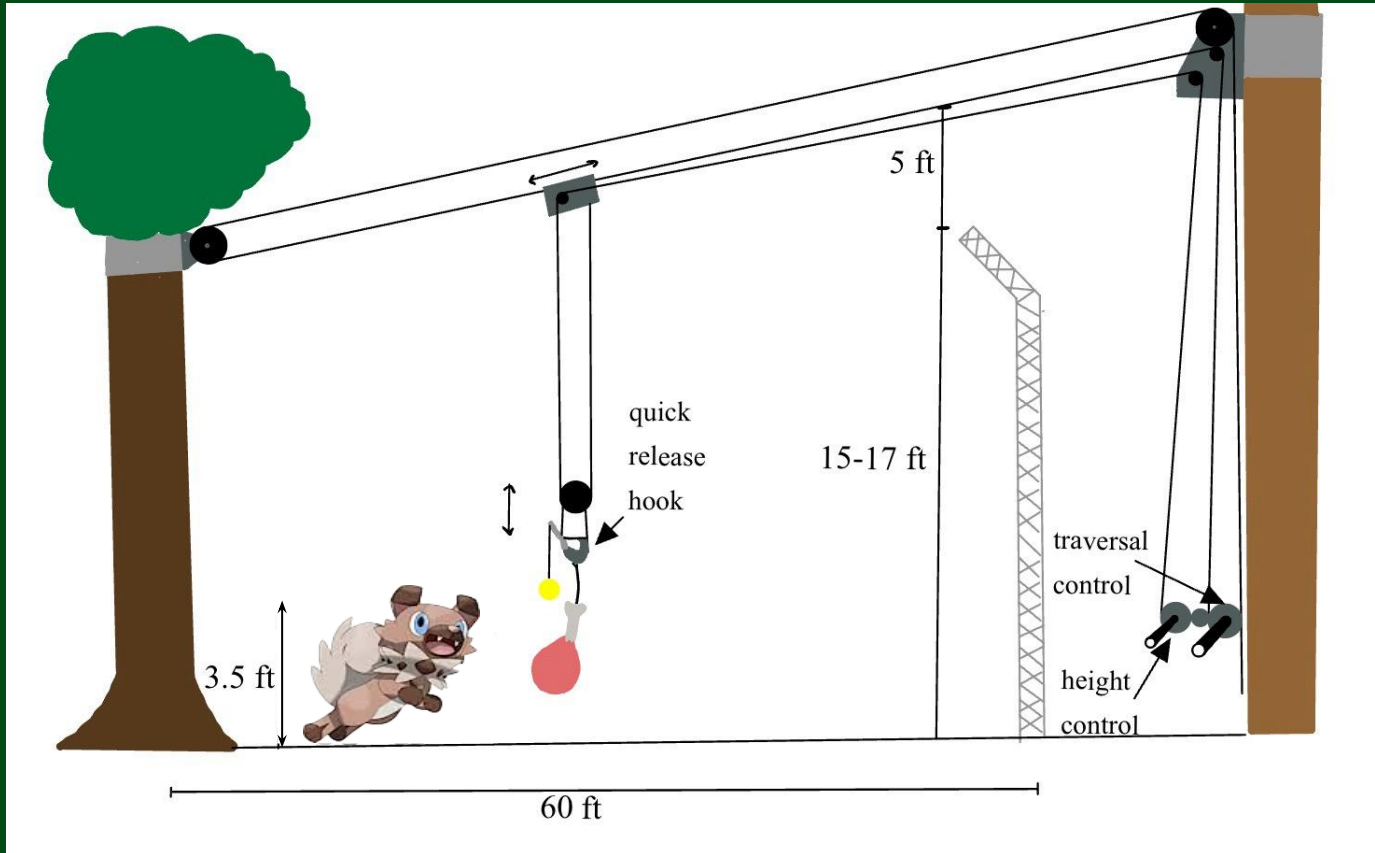
Concept	Score
Concept 1 (Meshed Gear Winch Cables)	141
Concept 2 (Counterweight)	161
Concept 3 (Ski Lift Cable)	134



Trade-Offs:

	Concept 1	Concept 2	Concept 3
Strengths	easy to control; easy to set up	Lower load; Mechanically simple	Simple operation
Weaknesses	Harder to reproduce; less resistant to climate	Unsure of feasibility as counterweight moves outwards	Heavy operator wear

Final Preliminary Concept



Future Plans

Key Decisions and Required Analysis:

- 1) Enclosure Location — Choose 1 of 14 enclosures
 - a) Worker input
 - b) Fence distance
 - c) Wolf enjoyment
- 2) Support Structure — Need to research materials
 - a) Support load
 - b) Resist weather
 - c) Easily installed
- 3) Wire Selection — Need to select material and gauge
 - a) Cost effective
 - b) Hold carcass
 - c) Withstand force of wolves



Cricket from Wolf Park

Category	Task	January	February	March	April	May
Construction	1st Prototype					
	To-Scale Prototype					
	Final Prototype Manufacture					
	Sub-assembly Demo					
	Final Prototype Assembly and Demo					
	Delivery					
	Cleanup					
Documentation	Concept Generation					
	PDR Slides/Present					
	PDR Documentation					
	BOM					
	CDR Slides/Present					
	FDR Slides/Present					
	Usage Instructions					
Analysis	FEA					
	Materials Analysis					
	To-Scale Prototype Testing					
	Final Prototype Testing					



Questions?



Donate to Wolf Park!



Save Wolves. Save Wilderness.